Subject: Physics

Difficulty: Easy

Topic: Kinematics

Question: What is the SI unit of velocity?

A) m/s

B) km/h

C) m/s^2

D) kg·m/s

Answer: A

Solution: Velocity is displacement divided by time, so its SI unit is metres per second (m/s).

Subject: Physics

Difficulty: Easy

Topic: Kinematics

Question: Which of these quantities has both magnitude and direction?

A) Speed

B) Displacement

C) Distance

D) Mass

Answer: B

Solution: Displacement is a vector. Speed, distance and mass are scalars.

Subject: Physics

Difficulty: Medium

Topic: Kinematics

Question: A car accelerates from rest at 2 m/s^2 for 5 s. How far does it travel?

A) 10 m

B) 50 m

C) 25 m

D) 5 m

Answer: C

Solution: s = ut + (1/2)at^2 = 0 + (1/2)(2)(5^2) = 25 m.

Subject: Physics

Difficulty: Medium

Topic: Dynamics

Question: What net force is needed to accelerate a 5 kg mass at 3 m/s^2?

A) 8 N

B) 1.67 N

C) 2 N

D) 15 N

Answer: D

Solution: Newton's second law: F = ma = 5 x 3 = 15 N.

Subject: Physics

Difficulty: Hard

Topic: Thermodynamics

Question: A Carnot engine works between 600 K and 300 K. What is its efficiency?

A) 50%

B) 25%

C) 75%

D) 100%

Answer: A

Solution: Efficiency = 1 - Tc/Th = 1 - 300/600 = 0.5, i.e. 50%.

Subject: Physics

Difficulty: Easy

Topic: Electricity

Question: What is the SI unit of electrical resistance?

A) Volt

B) Ohm

C) Ampere

D) Watt

Answer: B

Solution: Resistance is measured in ohms. Volt is potential difference, ampere is current, watt is power.

Subject: Physics

Difficulty: Medium

Topic: Electricity

Question: A 12 V battery is connected across a 4 ohm resistor. What current flows?

A) 48 A

B) 0.33 A

C) 3 A

D) 8 A

Answer: C

Solution: Ohm's law: I = V/R = 12/4 = 3 A.

Subject: Physics

Difficulty: Hard

Topic: Waves

Question: A wave has frequency 50 Hz and speed 340 m/s. What is its wavelength?

A) 0.147 m

B) 17000 m

C) 3.4 m

D) 6.8 m

Answer: D

Solution: v = f x wavelength, so wavelength = 340/50 = 6.8 m.

Subject: Mathematics

Difficulty: Easy

Topic: Algebra

Question: Solve for x: 2x + 6 = 14.

A) 4

B) 3

C) 5

D) 10

Answer: A

Solution: Subtract 6 from both sides: 2x = 8, so x = 4.

Subject: Mathematics

Difficulty: Easy

Topic: Algebra

Question: What is the value of 3^2 + 4^2?

A) 12

B) 25

C) 49

D) 14

Answer: B

Solution: 3^2 = 9 and 4^2 = 16, and 9 + 16 = 25.

Subject: Mathematics

Difficulty: Medium

Topic: Algebra

Question: What are the roots of x^2 - 5x + 6 = 0?

A) 1 and 6

B) -2 and -3

C) 2 and 3

D) 5 and 1

Answer: C

Solution: Factorise: (x - 2)(x - 3) = 0, so x = 2 or x = 3.

Subject: Mathematics

Difficulty: Medium

Topic: Calculus

Question: What is the derivative of x^3?

A) 3x

B) x^2

C) x^4/4

D) 3x^2

Answer: D

Solution: Power rule: d/dx x^n = n x^(n-1), so d/dx x^3 = 3x^2.

Subject: Mathematics

Difficulty: Hard

Topic: Calculus

Question: What is the value of the integral of 2x dx from 0 to 1?

A) 1

B) 0

C) 2

D) 1/2

Answer: A

Solution: The antiderivative of 2x is x^2. Evaluate from 0 to 1: 1 - 0 = 1.

Subject: Mathematics

Difficulty: Medium

Topic: Geometry

Question: What is the area of a circle of radius 7 (use pi = 22/7)?

A) 44

B) 154

C) 49

D) 22

Answer: B

Solution: Area = pi r^2 = (22/7) x 49 = 154.

Subject: Mathematics

Difficulty: Easy

Topic: Geometry

Question: What is the sum of the interior angles of a triangle?

A) 90 degrees

B) 270 degrees

C) 180 degrees

D) 360 degrees

Answer: C

Solution: The interior angles of any triangle always add up to 180 degrees.

Subject: Mathematics

Difficulty: Hard

Topic: Probability

Question: Two fair dice are rolled. What is the probability that the sum is 7?

A) 1/12

B) 1/9

C) 1/36

D) 1/6

Answer: D

Solution: Six of the 36 equally likely outcomes sum to 7, so P = 6/36 = 1/6.

Subject: Computer Science

Difficulty: Easy

Topic: Data Structures

Question: Which data structure follows the First-In-First-Out (FIFO) rule?

A) Queue

B) Stack

C) Tree

D) Graph

Answer: A

Solution: A queue removes the element that was added first. A stack is Last-In-First-Out.

Subject: Computer Science

Difficulty: Easy

Topic: Number Systems

Question: What is the decimal number 5 in binary?

A) 100

B) 101

C) 110

D) 111

Answer: B

Solution: 5 = 4 + 1 = 2^2 + 2^0, which is 101 in binary.

Subject: Computer Science

Difficulty: Medium

Topic: Algorithms

Question: What is the worst-case time complexity of binary search on a sorted array?

A) O(n)

B) O(n log n)

C) O(log n)

D) O(1)

Answer: C

Solution: Each step halves the search range, so at most log2(n) steps are needed.

Subject: Computer Science

Difficulty: Medium

Topic: Databases

Question: Which SQL command quickly removes all rows from a table but keeps the table itself?

A) DROP TABLE

B) DELETE TABLE

C) REMOVE TABLE

D) TRUNCATE TABLE

Answer: D

Solution: TRUNCATE TABLE empties the table and keeps its structure. DROP TABLE removes the table entirely.

Subject: Computer Science

Difficulty: Hard

Topic: Algorithms

Question: Why does Dijkstra's algorithm fail on graphs with negative edge weights?

A) It treats a node as final once it is the closest unvisited node, which a later negative edge can make wrong

B) It runs out of memory on large graphs

C) It cannot handle more than 100 nodes

D) It only works on trees

Answer: A

Solution: Dijkstra assumes that once a node is chosen its distance is final. A negative edge found later can give a shorter path, breaking that assumption.

Subject: Computer Science

Difficulty: Medium

Topic: Networking

Question: Which protocol is used to browse websites securely?

A) HTTP

B) HTTPS

C) FTP

D) SMTP

Answer: B

Solution: HTTPS is HTTP over TLS/SSL, so traffic between browser and website is encrypted.

Subject: Computer Science

Difficulty: Hard

Topic: Data Structures

Question: What is the height of a balanced binary search tree with n nodes?

A) O(n)

B) O(n^2)

C) O(log n)

D) O(1)

Answer: C

Solution: Balancing keeps the two sides about equal, so the height grows like log2(n).

Subject: Biology

Difficulty: Easy

Topic: Cell Biology

Question: Which organelle is known as the powerhouse of the cell?

A) Nucleus

B) Ribosome

C) Golgi body

D) Mitochondrion

Answer: D

Solution: Mitochondria release energy from glucose by respiration, producing most of the cell's ATP.

Subject: Biology

Difficulty: Easy

Topic: Cell Biology

Question: In which organelle does photosynthesis take place?

A) Chloroplast

B) Mitochondrion

C) Vacuole

D) Lysosome

Answer: A

Solution: Chloroplasts contain chlorophyll, which captures light energy for photosynthesis.

Subject: Biology

Difficulty: Medium

Topic: Genetics

Question: In a cross between two heterozygous plants (Tt x Tt), what percentage of offspring are expected to be tt?

A) 0%

B) 25%

C) 50%

D) 75%

Answer: B

Solution: The Punnett square gives TT, Tt, Tt, tt in the ratio 1:2:1, so tt is 1 in 4 = 25%.

Subject: Biology

Difficulty: Medium

Topic: Ecology

Question: In a food chain, which of these is a primary consumer?

A) Grass

B) Fox

C) Rabbit

D) Mushroom

Answer: C

Solution: Primary consumers eat producers. A rabbit eats grass, so it is a primary consumer.

Subject: Biology

Difficulty: Hard

Topic: Cell Biology

Question: A plant cell is placed in pure water. What happens?

A) Water leaves and the cell becomes plasmolysed

B) There is no net movement of water

C) The cell bursts because it has no cell wall

D) Water enters by osmosis and the cell becomes turgid

Answer: D

Solution: Pure water has a higher water potential than the cell, so water enters. The cell wall stops it bursting and the cell becomes firm (turgid).

Subject: Biology

Difficulty: Easy

Topic: Genetics

Question: What does DNA stand for?

A) Deoxyribonucleic acid

B) Dinucleic acid

C) Deoxyribose nucleic anhydride

D) Diribonucleic acid

Answer: A

Solution: DNA is short for deoxyribonucleic acid, the molecule that carries genetic information.

Subject: Biology

Difficulty: Medium

Topic: Genetics

Question: Which base pairs with adenine in DNA?

A) Guanine

B) Thymine

C) Cytosine

D) Uracil

Answer: B

Solution: In DNA, adenine always pairs with thymine (A-T) and guanine pairs with cytosine (G-C).